Grounding and Ground Fault Protection Considerations Specific to Emergency/Standby Generator Sets

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Introduction

Careful consideration of the grounding arrangements of AC generators used in emergency and standby power systems is essential for optimum continuity of power for critical loads and for the safety of personnel. There are grounding and ground fault protection considerations that are specific to emergency/standby generators when interconnected to a commercial AC power source through transfer switch equipment or when paralleled to another power source. Specific considerations for emergency and standby systems include selection of a system grounding method for the generator, when and when not to use four-pole transfer switch equipment, requirements for indication only of a ground fault on the generator, and the use of simultaneously switched neutral or overlapping the switched neutral.

System and Equipment Grounding

System and equipment grounding of generators follows the same set of guidelines in the National Electrical Code as for other AC power sources. The National Electrical Code defines the type of AC power systems to be grounded and the system grounding methods permitted, the circuit conductor to ground for various systems, the location of grounding connections, types and sizes of grounding conductors and electrodes, and methods of equipment grounding and bonding. AC system grounding is the intentional connection between a conductor of an AC power system and ground. Equipment grounding is the bonding and grounding of all non-current carrying (during normal operation) metal conduit, equipment enclosures, supports, frames, etc. used in the distribution of AC power.

Applications such as telecommunications and data processing, with circuits used for digital data transmission at low energy levels, typically have equipment grounding practices that supplement those in the NEC. There should be no conflict between equipment grounding for safety as required by the NEC and effective grounding for sensitive electronic equipment.

Selection of System Grounding Method

There are several types of system grounding typically in use for low voltage (600 volts and below) AC power systems. Three-phase electrical systems may be solidly grounded, high-impedance grounded (usually with a grounding resistor), or ungrounded. The system grounding method chosen for the generator does not necessarily have to be the same as used on the commercial power supply. The National Electrical Code, Article 250-5, requires AC power systems of 50-1000 V to be solidly grounded where the maximum voltage to ground on the ungrounded conductors will not exceed 150 V. Single-phase systems and three-phase systems that supply phase-to-neutral loads are also required to be solidly grounded. Three-phase three-wire AC power systems that are not required to be solidly grounded may be either solidly grounded, high-impedance grounded, or ungrounded. See Figure 1A. AC power systems are high-impedance grounded or ungrounded if it is desirable to limit the magnitude of ground fault current or minimize the interruption of power due to phase-to-ground faults.

Most typical low voltage emergency standby systems are required to serve line-to-neutral loads. The generator windings are wye-connected, the neutral is brought out for use as a circuit conductor and the generator is solidly-grounded. When the neutral is being used as a circuit conductor the generator would not be permitted to be high-resistance grounded. However, where line-to-neutral loads need not be served, or where delta-wye isolation transformers are used to derive a neutral for L-N loads, a 3-phase 3-wire generator may be high resistance grounded. Often the percentage of L-N emergency loads is low relative to the capacity of the generator set so that the cost of transformers to derive a neutral may be offset by gains in service continuity.

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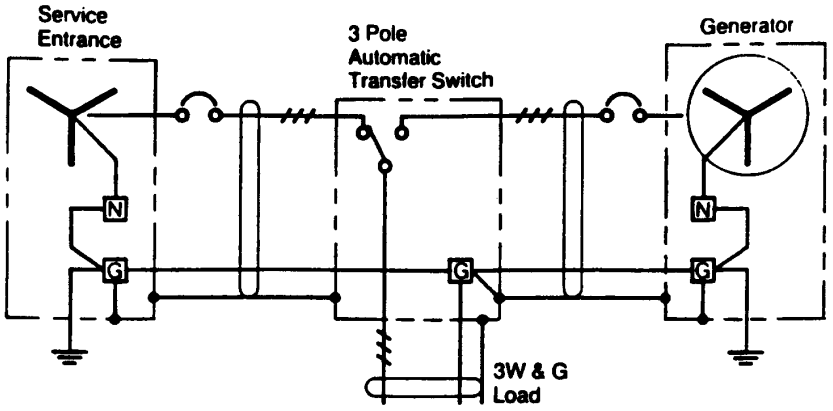
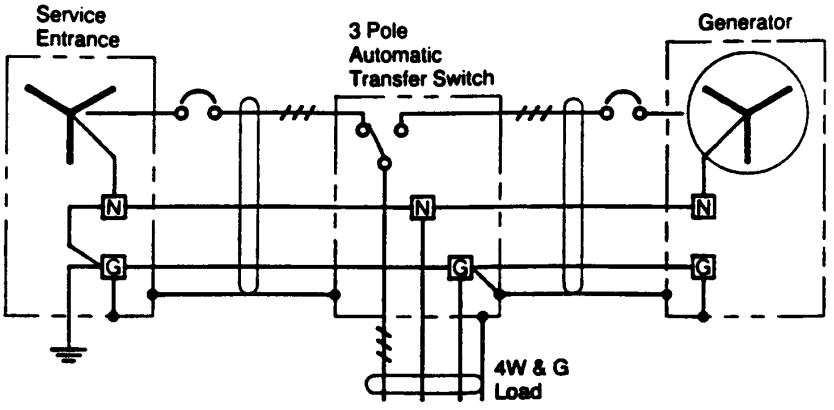
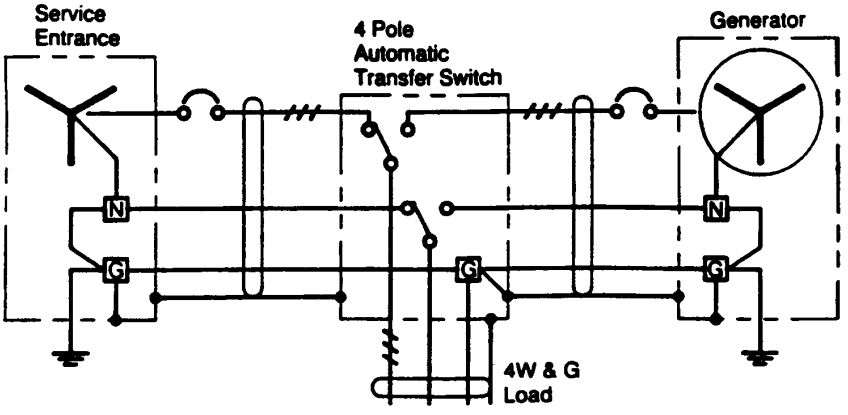


Figure 1A. Separately–derived Generator (No Neutral)



1B. Non–Separately–derived Generator



1C. Separately–derived Generator (With Neutral)

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Where the normal commercial AC service is solidly grounded, and the generator neutral is interconnected with the neutral from the normal service at the neutral terminal block in 3-pole transfer equipment (Figure 1B), the generator AC system is considered effectively grounded. In this case the generator would not be considered a separately derived system, and the generator neutral must NOT be bonded to the generator frame (equipment ground) or GFP sensing problems may result. The National Electrical Code makes this clear. NEC Article 250-5 (d) (FPN No. 1): "An alternate alternating current power source such as an on-site generator is not a separately derived system if the neutral is solidly interconnected to a service-supplied system neutral." Where the transfer equipment includes a switched neutral pole, there is not a solid interconnection with the service-supplied neutral, so the generator is a separately derived system and its neutral must be grounded in accordance with NEC 250-26. See Figure 1C.

Switched Neutral Transfer Switch Equipment

Transfer switch equipment for three phase systems is available as 3-pole with a neutral terminal block or 4-pole where the fourth pole is used to switch the neutral. Generally, whenever a zero-sequence-type (core balance) sensing function for ground fault current is used on either source, or may be in the future, 4-pole transfer switches and grounding the generator neutral locally as a separately-derived system are recommended. If core balance ground fault sensing for indication (alarm) or protection is provided on the generator, 4-pole transfer equipment will be necessary
for proper sensing.

Depending on the manufacturer, the switched neutral pole may be either simultaneously switched with the phase poles from source to source, or with an overlapping action of the neutral contacts. Simultaneously switched neutral pole transfer equipment does not connect together the grounded neutrals of the commercial AC power source and the separately-derived generator. Overlapping neutral contacts momentarily connect the grounded neutrals of both the commercial AC source and the AC generator source while the phase poles switch simultaneously. When the phase poles have transferred the neutral contact on the open source also opens. Multiple neutral-ground loops exist during the overlap period of the neutral contacts. If a potential difference exists between the commercial AC power system ground and the generator ground, the uncontrolled flow of current over the grounding system may cause "noise" problems for sensitive electronic equipment.

Switched Neutral Required/Indicated

There are some specific cases where it will be necessary to use 4-pole transfer switches or cases where a 4-pole transfer switch should be considered. Descriptions of these specific cases follow.

Case 1. AC Generator with Ground Fault Sensing

If an emergency generator set is rated 277/480 or 347/600 solidly-grounded wye, and the generator main disconnecting means is rated 1000 amps or more, NEC Article 700-7 (d) will require a ground fault indication on the generator (figure 2). This NEC requirement does not apply to generators that are classified as legally required standby or optional standby. Note that this requirement is for an indication (alarm) only and ground fault protection is not required. Circuit interruption would result in a loss of power to emergency systems involving life safety. While ground fault indication is required by NEC on life safety generators above 1000 amps, it can be and often is provided below 1000 amps. Ground fault protection of equipment may be provided on generators in other applications, where the equipment protection is more important than the continuity of generator power.

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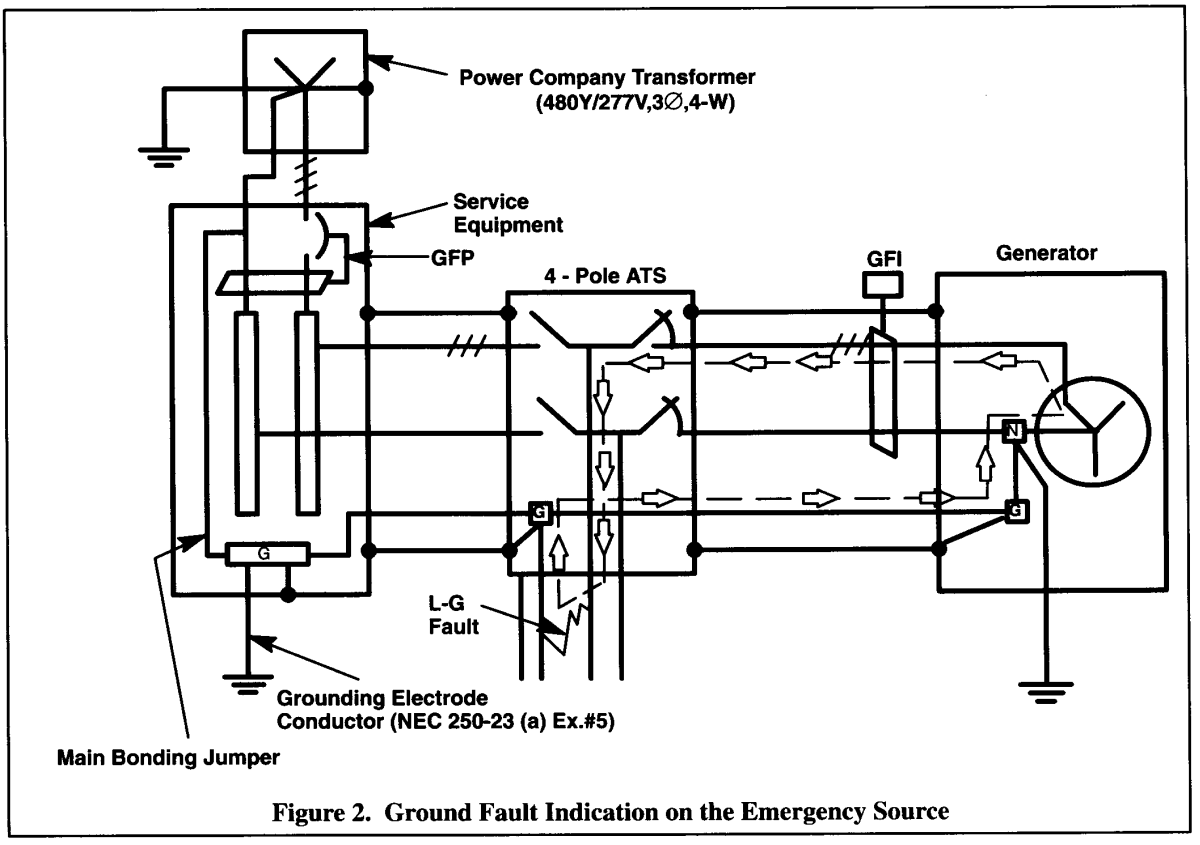


Figure 2. Ground Fault Indication on the Emergency Source

Sometimes the NEC requirement of 230-95 for ground fault protection on high capacity services is interpreted to apply to emergency standby generators as well. However, the NEC states in 700-26 that ground fault protection of emergency services, which would include a trip, shall not be required. Providing ground fault protection which includes circuit interruption may be contrary to the intent of NFPA standards for essential electrical systems. The NFPA standards suggest that higher priority be given to continuity of service than to the protection of essential electrical system equipment, except where equipment protection is required to prevent a greater hazard than lack of essential electrical service. In general, it is not recommended to provide ground fault protection which includes tripping off critical emergency power.

Case 2. Two-levels of GFP and Two or more Transfer Switches.

Figure 3 illustrates a potential problem with the arrangement where the normal distribution has GFP on the service and GFP on the feeders, and the neutral conductors are solidly interconnected in two or more transfer switches. In this case parallel paths exist for neutral current caused by unbalanced loading or third-harmonic non-linear loads, so that the neutral current may split according to the impedances of the parallel paths. The second level GFP sensor on the feeder may be bypassed such that there is a unbalanced current sensed at both feeder GFP sensors, which could result in a nuisance trip of either or both feeders. Parallel paths for ground fault current may also be the result of multiple ground faults which could defeat the GFP sensors.

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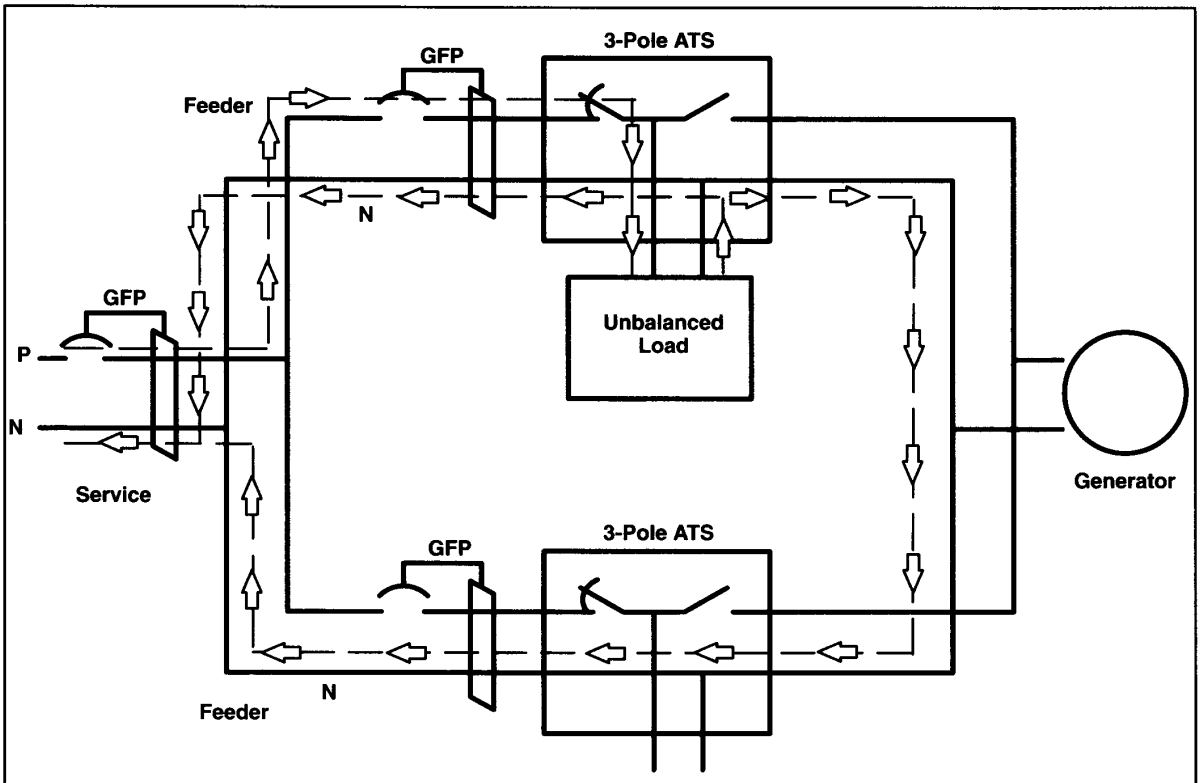


Figure 3. Multiple Paths for Neutral Current

Case 3. GFP on Normal Service, Transfer Switch Fed by Generator.

The operation of the GFP equipment must also be considered when normal power is available and the generator source connected to the load as it would be during tests and during time delays on retransfer as shown in Figure 4. Because the generator neutral is not grounded, the only path for ground fault current back to the source is through the equipment grounding conductor to the normal service equipment, through the GFP sensor, and back to the generator on the neutral. If a ground fault occurs while normal power is available and the generator is supplying power to the load, a potential exists for a trip of the GFP equipment at the utility service. If this possibility were to occur, the transfer switch control would not allow a retransfer operation to an open disconnect even though utility power was available on the line side. The limited circumstances under which this could occur and the extent to which this would be a problem warrants engineering judgement to determine whether to use 4-pole transfer switches and ground the generator as a separately-derived system.

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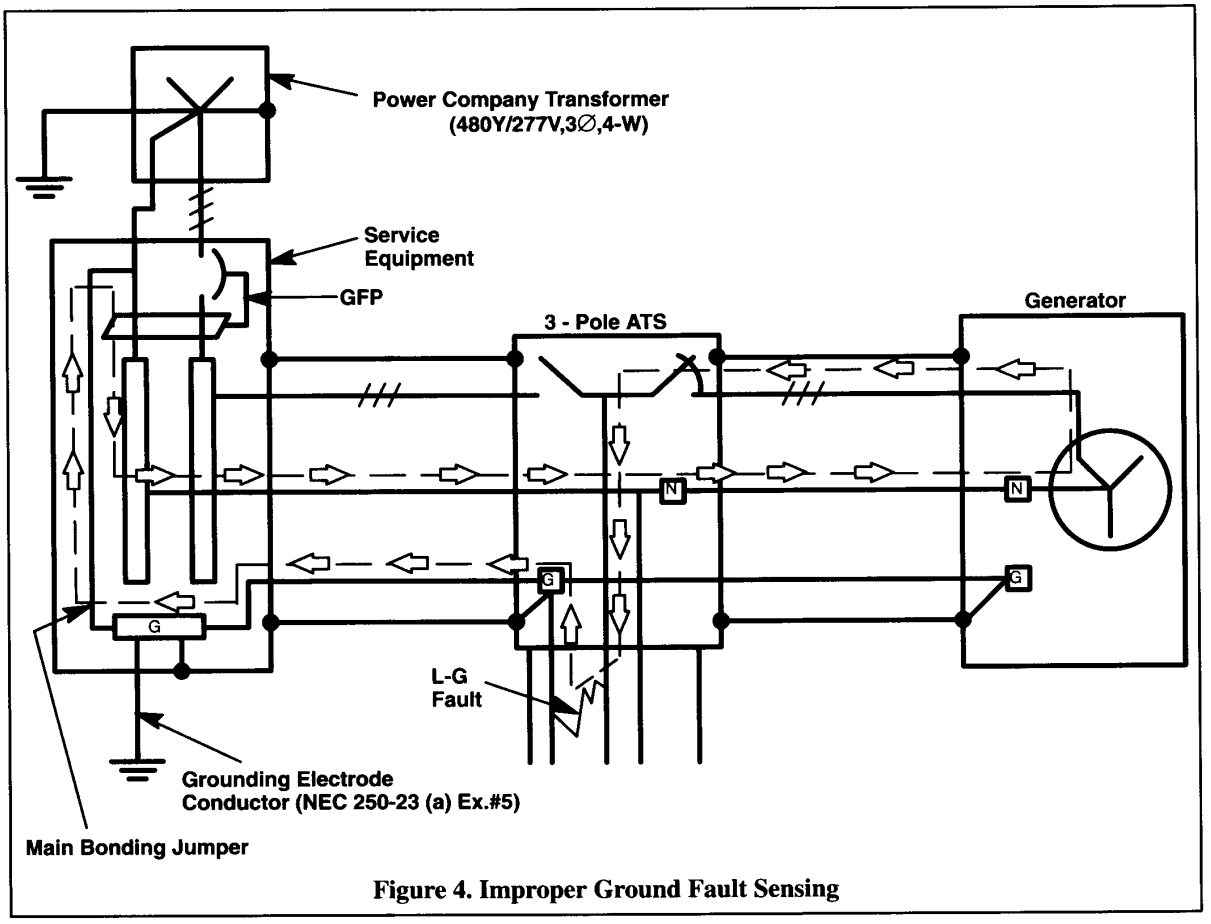


Figure 4. Improper Ground Fault Sensing

Case 4. Grounding Sign NEC Requirement

A new NEC requirement calls for a sign to be placed at the commercial AC service equipment where the generator neutral is grounded on systems using 3-pole transfer switches as shown in Figure 5. The sign must specify that the generator is grounded at that location. If the main bonding jumper is disconnected from the neutral at the commercial AC source service equipment while the generator set is running, a voltage potential exists between the neutral and ground. When the operating generator's neutral is disconnected from ground the neutral "floats" at a potential to ground. If a 4-pole transfer switch were to be substituted in the diagram for the 3-pole switch and the generator grounded as a separately-derived source, the fourth pole would open the neutral between the generator and the service equipment eliminating the potential.

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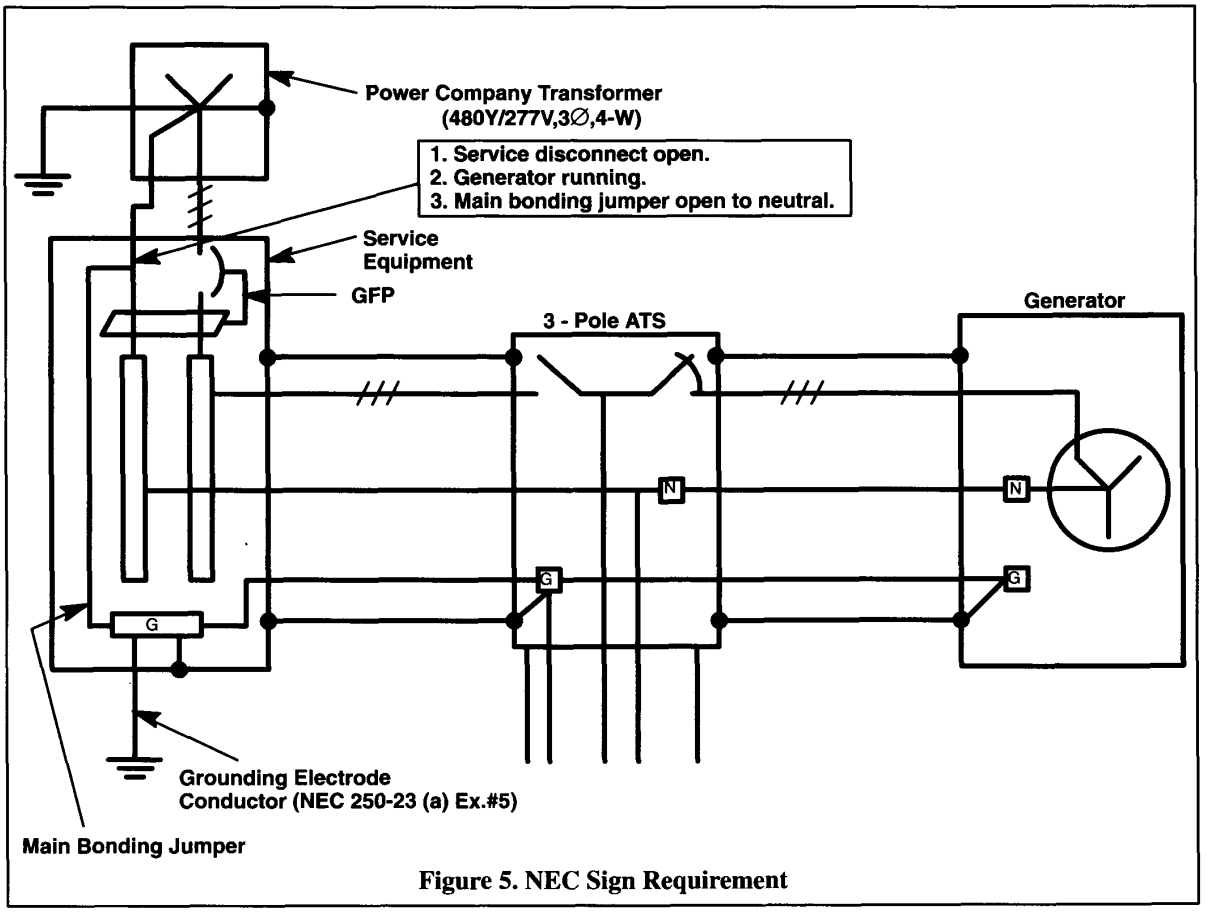


Figure 5. NEC Sign Requirement

Switched Neutral Not Recommended

There is also a case where the use of switched neutral transfer equipment would not be recommended. This case is static UPS equipment where the UPS inverter neutral is grounded on the commercial AC service side of the automatic transfer switch, as shown in Figure 6A. The UPS bypass circuit uses the grounded neutral of the service and connects directly to the wye of the inverter output so that the UPS is not grounded locally as a separately-derived source. Transferring the neutral of this type of UPS equipment may interrupt the sensing and monitoring of the UPS. Where the bypass input of the UPS requires a grounded-wye feeder, use of a bypass transformer as shown in Figure 6B on the load side of the transfer switch permits use of a 3-phase 3-wire feeder and 3-pole transfer switch. Refer to IEEE Recommended Practice for Grounding of Industrial and Commercial Power Systems, IEEE Std 142-1991 for additional information.

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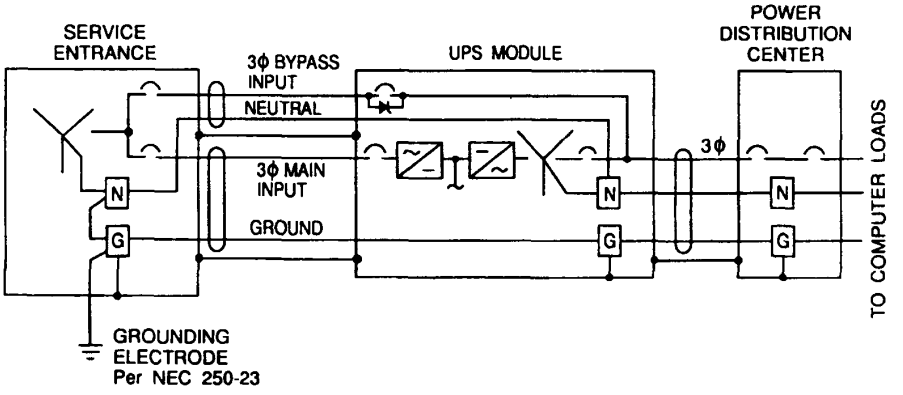


Figure 6A. Non–isolated Bypass Neutral

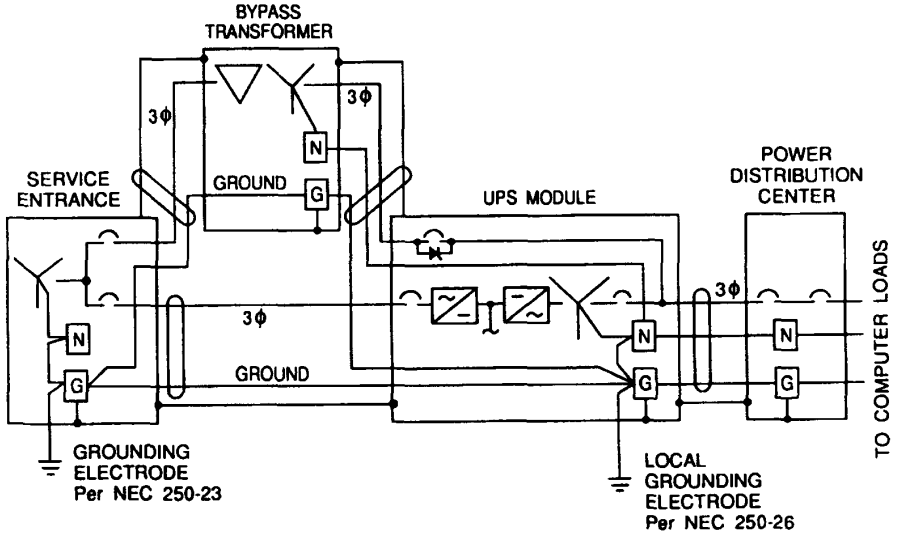


Figure 6B. Isolated Bypass Neutral

Summary

System and equipment grounding of generators follows the same set of guidelines in the National Electrical Code as for other AC power sources. When the neutral is used as a circuit conductor with a 3-pole transfer switch the generator is not a separately-derived source. When the neutral is used as a circuit conductor with a 4-pole transfer switch the generator is a separately-derived source. Four-pole transfer switches may be required for proper sensing of ground fault protection or indication equipment.

References

1. NFPA 70-1993, National Electrical Code, National Fire Protection Association, Inc., Quincy MA.
2. IEEE Std 142-1991, IEEE Recommended Practice for Grounding of Industrial and Commercial Power Systems, Institute of Electrical and Electronics Engineers, Inc., New York NY.
3. IEEE Std 446-1987, IEEE Recommended Practise for Emergency and Standby Power Systems for Industrial and Commercial Applications, Institute of Electrical and Electronics Engineers, Inc., New York NY.

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